Differentially Private Accelerated Distributed Algorithm for Aggregative Optimization

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***Abstract*—This article studies the distributed aggregative optimization (DAO) problem, wherein each agent's local objective function depends not only on its own decision variables but also on an aggregate term involving all agents' decisions. In such settings, frequent information exchange among agents raises serious privacy concerns, as sensitive information may be inferred from shared data. To address this issue, we propose a differentially private accelerated distributed gradient tracking algorithm that integrates techniques from distributed dynamic average consensus, the heavy-ball momentum method, and differential privacy (DP). Specifically, to preserve privacy, the exchanged information is perturbed with independent Laplace noise. Moreover, our algorithm uses a noise deduction mechanism to prevent the accumulation of errors caused by noise during the estimation of aggregate variables and local gradients, thereby ensuring the algorithm's accuracy. Under the assumption that the global objective function is strongly convex and has Lipschitz-continuous gradients, we rigorously prove that the proposed algorithm achieves linear convergence in the mean-square error sense. In addition, we derive explicit suboptimality bounds and formally establish that the algorithm satisfies ϵ-DP. Finally, numerical simulations are provided to validate the effectiveness of the proposed method.**

***Index Terms*—Differential privacy (DP), distributed aggregative optimization (DAO), gradient tracking, heavy-ball momentum method.**

I. Introduction

In recent years, distributed optimization has received widespread attention across various fields [1], [2], [3]. This paradigm addresses problems where the global objective function is the sum of local objective functions, each depending only on its own decision variables. The objective is to collaboratively minimize the global objective through local computations and interagent communication. Numerous distributed optimization algorithms have been proposed and have demonstrated excellent results, such as gradient-based methods [4], [5], distributed alternating direction method of multipliers (ADMM) [6], [7], and decentralized exact first-order algorithms [8]. To accelerate convergence, a variety of momentum-based techniques have been incorporated into gradient tracking frameworks, such as the heavy-ball method [9], Nesterov's momentum method [10], and other momentum-enhanced schemes [11], [12]. These algorithms have been shown to achieve linear convergence for smooth and strongly convex objective functions, offering significant improvements in convergence speed over nonaccelerated methods.

It is important to note that most of the aforementioned algorithms focus on consensus optimization problems, where agents aim to agree on a common decision variable. However, in many practical applications, such as optimal sensor localization [13], multirobot surveillance [14], and energy management in smart grids [15], each agent's local objective depends not only on its own decision variables but also on an aggregate of all the agents' decisions. This class of problems is referred to as distributed aggregative optimization (DAO) [13], and it generalizes consensus optimization by incorporating coupling through aggregate terms.

The study on DAO problem originates from [13], where the distributed aggregative gradient tracking (DAGT) algorithm is proposed for multiagent systems. Building on this foundational work, a series of subsequent studies [16], [17], [18], [19], [20] develop various constrained DAO algorithms. Specifically, [16] introduces the online distributed gradient tracking (O-DGT) algorithm and establishes a three-component upper bound structure for the dynamic regret. Wang and Yi [17] propose a gradient-tracking-based distributed Frank–Wolfe algorithm that replaces projection operations with linear optimization, thereby efficiently solving convex DAO problems over time-varying networks. To address DAO problems with coupled affine constraints, Carnevale et al. [18] propose a primal–dual algorithm that incorporates dual diffusion and gradient tracking mechanisms and prove its linear convergence rate. Li et al. [19] present a projected aggregation tracking algorithm with constant step sizes, which not only achieves a constant bound on dynamic regret but also demonstrates linear convergence under static conditions. Meanwhile, [20] extends the DAO methodology to high-order nonlinear systems. Huang et al. [21] investigate the formation control problem for multirobot systems subject to local constraint sets. Furthermore, Chen et al. [22] introduce a communication-efficient algorithm that uses compression techniques for solving DAO problems. To achieve acceleration, [23], [24], [25] integrated various acceleration strategies into the DAGT framework.

However, a critical challenge in distributed algorithms lies in the frequent information exchange among agents, which inherently increases the risk of privacy breaches. Communication between agents can be intercepted by adversaries who may exploit compromised links to access transmitted data. By leveraging advanced inference techniques, such

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adversaries can potentially extract sensitive information from individual agents. For instance, Mandal [26] presents a theoretical analysis of privacy leakage in distributed optimization, demonstrating how eavesdroppers in economic dispatch problems can accurately infer cost function parameters and power generation outputs. Similarly, Wang et al. [27] investigate privacy vulnerabilities in aggregative games, where the cost functions often involve market prices or marginal cost parameters. In such scenarios, if adversaries gain access to gradients and strategies, they may reconstruct players' private cost structures and operational decisions, potentially leading to significant economic losses for the participants.

As a result, privacy preservation has become a pressing concern in the design of distributed algorithms. Among various privacy-preserving techniques, differential privacy (DP) has gained significant attention due to its rigorous theoretical foundation, strong privacy guarantees, and ease of implementation [28], which distinguish it from encryption-based approaches [29] and state-decomposition-based methods [30]. Recent works have explored the application of DP in both distributed optimization [31], [32] and aggregative games [33], [34]. Specifically, the DiaDsp algorithm proposed in [31] incorporates gradient tracking along with decaying Laplacian noise injected into interagent communications to achieve ϵ-DP. Zhang et al. [32] develop a differentially private consensus algorithm based on dynamic encoding–decoding mechanisms for second-order multiagent systems. In the context of aggregative games, [33] protects cost function parameters by injecting Laplacian noise into the transmitted data and further analyzes the tradeoff between convergence accuracy and privacy levels. In addition, Wang and Nedić [34] propose a distributed equilibrium computation approach that achieves both rigorous DP and accurate computation of the Nash equilibrium.

Despite the growing body of research, little attention has been paid to incorporating DP into DAO problems. To the best of our knowledge, privacy-preserving methods that exploit momentum-based acceleration within the DAO framework remain unexplored. To fill this gap, this article develops a novel algorithm that integrates DP mechanisms with momentum-based acceleration for DAO. The main contributions are summarized as follows.

1) We propose a privacy-preserving heavy-ball DAGT algorithm (PP-HB-DAGT) for DAO problems. The proposed algorithm integrates gradient tracking and heavy-ball momentum while ensuring DP by injecting carefully calibrated Laplace noise into exchanged messages. To mitigate noise accumulation caused by the interaction between momentum acceleration and aggregative dynamics, a dedicated noise deduction mechanism is introduced.
2) We establish rigorous convergence guarantees under constant step sizes. It is shown that the proposed algorithm achieves linear convergence to a neighborhood of the optimal solution, and an explicit bound on the steady-state mean-square error is derived.
3) We provide a unified convergence and DP analysis tailored to the DAO setting. The analysis quantitatively characterizes the tradeoff between convergence accuracy and privacy levels, which has not been previously established for DAO problems. Comparative simulations demonstrate that the proposed algorithm achieves superior accuracy, while providing strong privacy guarantees.

The remainder of this article is structured as follows. Section II introduces the necessary preliminaries and formulates the DAO problem. Section III presents the proposed algorithm along with several essential supporting lemmas. Sections IV and V provide rigorous analyses of the algorithm's convergence performance and privacy guarantees, respectively. Section VI conducts numerical simulations to validate the theoretical findings. Finally, Section VII concludes this article.

Notations: Let $\mathbb{R}$ denote the set of real numbers. $\mathbb{R}^n$ and $\mathbb{R}^{n\times m}$ represent the sets of n-dimensional real-valued column vectors and $n \times m$-dimensional real-valued matrices, respectively. Define $\mathrm{col}\{x_1,\dots,x_n\} = [x_1^T,\dots,x_n^T]^T$ to represent the column vector formed by stacking $x_1, x_2, \dots, x_n$. Let $\mathbf{1}_n$ and $\mathbf{0}_n$ denote the n-dimensional column vectors with all elements equal to 1 and 0, respectively, and let I denote the identity matrix of appropriate size. For a vector x, x_i denotes its ith element. Let $\mathrm{Lap}(\theta)$ represent the Laplace distribution with probability density function $f_L(x) \triangleq (1/2\theta)e^{-|x|/\theta}$. This distribution has zero mean and variance $2\theta^2$. In addition, for any $x, y \in \mathbb{R}$, we have $(f_L(x)/f_L(y)) \leq e^{(|y-x|/\theta)}$. In addition, $\mathbb{E}(\Phi)$ denotes the expectation of Φ. Let $\rho(A)$ denote the spectral radius of a square matrix A, $\|\cdot\|$ denote the standard Euclidean norm, and $\|\cdot\|_1$ denote the 1-norm. The Kronecker product is represented by $\otimes$.

II. Preliminaries and Problem Formulation

A. Graph Theory

We consider a network of N agents that communicate over a weighted undirected graph denoted by $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, A\}$, where $\mathcal{V} = \{1,\dots,N\}$ is the agent set, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set, and $A = [a_{ij}] \in \mathbb{R}^{N\times N}$ is the weighted adjacency matrix. The weight $a_{ij} > 0$ if $(j,i) \in \mathcal{E}$, implying that agent j can transmit information directly to agent i, and $a_{ij} = 0$ otherwise. The neighbor set of agent i is defined as $\mathcal{N}_i = \{j \in \mathcal{V} : (j,i) \in \mathcal{E}\}$. The graph is said to be connected if there exists a path between any pair of distinct agents $i, j \in \mathcal{V}$. We make the following standard assumption regarding the communication graph.

Assumption 1: The communication topology $\mathcal{G}$ is undirected and connected. The associated adjacency matrix $A = [a_{ij}] \in \mathbb{R}^{N\times N}$ is nonnegative and doubly stochastic, i.e., $\mathbf{1}_N^T A = \mathbf{1}_N^T$ and $A\mathbf{1}_N = \mathbf{1}_N$.

Under Assumption 1, it follows that $\mathcal{J}\mathcal{A} = \mathcal{A}\mathcal{J} = \mathcal{J}$, where $\mathcal{A} = A \otimes I_d$, $\mathcal{J} = J \otimes I_d$, and $J = (1/N)\mathbf{1}_N\mathbf{1}_N^T$.

B. Problem Formulation

Consider the following DAO problem:

$$\min_{\boldsymbol{x}\in\mathbb{R}^{Nq}} F(\boldsymbol{x}) = \sum_{i=1}^{N} f_i(x_i, \sigma(\boldsymbol{x})) \tag{1}$$

where $f_i : \mathbb{R}^q \times \mathbb{R}^d \to \mathbb{R}$ is the local objective function of agent i, and $\boldsymbol{x} = \mathrm{col}\{x_1,\dots,x_N\} \in \mathbb{R}^{Nq}$ is the global decision variable, with $x_i \in \mathbb{R}^q$ being the local decision variable of agent i. The term $\sigma(\boldsymbol{x}) = (1/N)\sum_{i=1}^N \phi_i(x_i)$ represents the aggregative function that couples all the agents through their decision variables, and $\phi_i : \mathbb{R}^q \to \mathbb{R}^d$ is a local function known only to agent i.

For brevity, we let $\nabla_1 f_i(x_i, \sigma(\boldsymbol{x}))$ and $\nabla_2 f_i(x_i, \sigma(\boldsymbol{x}))$ denote the gradients of f_i with respect to its first and second arguments, respectively. Define $f(\boldsymbol{x},\boldsymbol{\sigma}) = \sum_{i=1}^{N} f_i(x_i,\sigma_i) : \mathbb{R}^{Nq} \times \mathbb{R}^{Nd} \to \mathbb{R}$, where $\sigma_i \in \mathbb{R}^d$, $\boldsymbol{\sigma} = \mathrm{col}\{\sigma_1,\ldots,\sigma_N\} \in \mathbb{R}^{Nd}$, and

$$\nabla\phi(\boldsymbol{x}) := \begin{bmatrix} \nabla\phi_1(x_1) & & \\ & \ddots & \\ & & \nabla\phi_N(x_N) \end{bmatrix} \in \mathbb{R}^{Nq\times Nd}.$$

Then, the gradient of the global objective function $F(\boldsymbol{x})$ can be expressed as $\nabla F(\boldsymbol{x}) = \nabla_1 f(\boldsymbol{x}, \mathbf{1}_N \otimes \sigma(\boldsymbol{x})) + \nabla\phi(\boldsymbol{x}) \left[\mathbf{1}_N \otimes (1/N)\sum_{i=1}^{N} \nabla_2 f_i(x_i,\sigma(\boldsymbol{x}))\right]$, where $\nabla_2 f_i(x_i,\sigma(\boldsymbol{x})) = \nabla_2 f_i(x_i,\sigma_i)|_{\sigma_i=\sigma(\boldsymbol{x})}$.

We impose the following assumption on the objective functions.

Assumption 2: The objection functions in (1) satisfy the following conditions.

1) The global objective function $F(\boldsymbol{x})$ is differentiable and μ-strongly convex. In addition, $F(\boldsymbol{x})$ is L_1-smooth, namely, $\nabla_1 f(\boldsymbol{x},\boldsymbol{\sigma}) + \nabla\phi(\boldsymbol{x})\left[\mathbf{1}_N \otimes (1/N)\sum_{i=1}^{N} \nabla_2 f_i(x_i,\sigma_i)\right]$ is L_1-Lipschitz continuous.
2) For each $i \in \mathcal{V}$, the local gradient $\nabla_2 f_i(x_i,\sigma_i)$ is $L_{2,i}$-Lipschitz continuous. That is, there exists a constant $L_{2,i} > 0$ such that for all $x_{i,1}, x_{i,2} \in \mathbb{R}^q$, $\sigma_{i,1},\sigma_{i,2} \in \mathbb{R}^d$, $\|\nabla_2 f_i(x_{i,1},\sigma_{i,1}) - \nabla_2 f_i(x_{i,2},\sigma_{i,2})\| \le L_{2,i}(\|x_{i,1} - x_{i,2}\| + \|\sigma_{i,1} - \sigma_{i,2}\|)$.
3) For each $i \in \mathcal{V}$, the local function $\phi_i(x_i)$ is differentiable. There exists a constant $L_{3,i} > 0$ such that for all $x_i \in \mathbb{R}^q$, $\|\nabla\phi_i(x_i)\| \le L_{3,i}$.

Assumption 2- 1) is standard in the distributed optimization literature [7], [9], [13], [14] and ensures the existence and uniqueness of the global minimizer $\boldsymbol{x}^*$, satisfying $\nabla F(\boldsymbol{x}^*) = \mathbf{0}_{Nq}$. From Assumption 2–2), it follows that $\nabla_2 f(\boldsymbol{x},\boldsymbol{\sigma})$ is L_2-Lipschitz continuous with $L_2 = \max_{i\in\mathcal{V}}\{L_{2,i}\}$. Similarly, Assumption 2–3) implies that $\|\nabla\phi(\boldsymbol{x})\| = \max_{i\in\mathcal{V}}\|\nabla\phi_i(x_i)\| \le \max_{i\in\mathcal{V}}\{L_{3,i}\} \triangleq \hat{L}_3$, and we can get the fact that $\|\phi_i(x_{i,1}) - \phi_i(x_{i,2})\| \le L_{3,i}\|x_{i,1} - x_{i,2}\|$.

C. Differential Privacy

DP provides a rigorous mathematical framework to quantify privacy guarantees in statistical computations. It ensures that the output of an algorithm remains nearly unchanged when a single individual's data are modified, thereby limiting the ability of adversaries to infer sensitive information. For a more comprehensive understanding of DP, the readers are referred to [28]. To protect the decision variables, we first define adjacency and DP as follows.

Definition 1 (Adjacency): Consider two different decision sets $\Psi^{(1)} = \{x_i^{(1)}\}_{i=1}^N$ and $\Psi^{(2)} = \{x_i^{(2)}\}_{i=1}^N$. These two decision sets $\Psi^{(1)}$ and $\Psi^{(2)}$ are said to be δ-adjacent if there exists $i_0 \in \mathcal{V}$ and a given constant $\delta > 0$ such that

$$x_i^{(1)} = x_i^{(2)} \quad \forall i \neq i_0, \text{ and } D\left(x_{i_0}^{(1)}, x_{i_0}^{(2)}\right) = \|x_{i_0}^{(1)} - x_{i_0}^{(2)}\| \le \delta.$$

Definition 2 (Differential Privacy): An iterative algorithm $\mathcal{M}$ is said to satisfy ϵ-DP if, for any two δ-adjacent variable sets $\Psi^{(1)}$ and $\Psi^{(2)}$, and for any measurable set $\mathcal{H} \subseteq \mathrm{Range}(\mathcal{M})$, the following holds:

$$P\left\{\mathcal{M}\left(\Psi^{(1)}\right) \in \mathcal{H}\right\} \le e^{\epsilon} P\left\{\mathcal{M}\left(\Psi^{(2)}\right) \in \mathcal{H}\right\} \tag{2}$$

where $\mathrm{Range}(\mathcal{M})$ is the domain of observable outputs under algorithm $\mathcal{M}$, and $\mathcal{H}$ is the set of possible observations, that is, the information transmitted over the communication network. The details of $\mathcal{H}$ will be elaborated later.

Remark 1: Definition 2 explains the essence of DP. If the outputs of an algorithm are statistically indistinguishable for any pair of adjacent datasets, then an adversary observing the algorithm's output cannot reliably infer which dataset was used as input. Consequently, the private information of individual agents is protected. The parameter $\epsilon > 0$, referred to as the privacy budget, quantifies the level of privacy, and a smaller value of ϵ indicates stronger privacy guarantees.

III. Algorithm Design and Supporting Lemmas

In this section, a DAO algorithm with DP is proposed, and several supporting lemmas used for the convergence and privacy analyses are provided.

A. Algorithm Design

To address the DAO problem formulated in (1), we integrate the DAGT algorithm [13] with the heavy-ball momentum method and propose the following update rules:

$$\begin{aligned} x_{i,k+1} = x_{i,k} - \alpha\left[\nabla_1 f_i\left(x_{i,k},\sigma_{i,k}\right) + \nabla\phi_i\left(x_{i,k}\right)y_{i,k}\right] \\ + \beta\left(x_{i,k} - x_{i,k-1}\right) \end{aligned} \tag{3a}$$

$$\sigma_{i,k+1} = \sum_{j=1}^{N} a_{ij}\sigma_{j,k} + \phi_i\left(x_{i,k+1}\right) - \phi_i\left(x_{i,k}\right) \tag{3b}$$

$$\begin{aligned} y_{i,k+1} = \sum_{j=1}^{N} a_{ij}y_{j,k} + \nabla_2 f_i\left(x_{i,k+1},\sigma_{i,k+1}\right) \\ - \nabla_2 f_i\left(x_{i,k},\sigma_{i,k}\right) \end{aligned} \tag{3c}$$

where each agent $i \in \mathcal{V}$ locally maintains auxiliary variables $\sigma_{i,k}$ and $y_{i,k}$ to track the aggregative term $\sigma(\boldsymbol{x})$ and the average gradient $(1/N)\sum_{i=1}^{N}\nabla_2 f_i(x_i,\sigma(\boldsymbol{x}))$, respectively. The update rules in (3b) and (3c) are inspired by the dynamic average consensus protocol in [35].

Although agents do not share raw data directly, privacy leakage can still occur during iterative updates and information exchange. As shown in [27], an eavesdropper may infer sensitive local data by analyzing transmitted information. To address this issue, we enhance the algorithm (3) by incorporating DP mechanisms, yielding the PP-HB-DAGT algorithm, as summarized in Algorithm 1. Specifically, each agent i injects independent noise $\eta_{i,k} = \mathrm{col}\{\eta_{i1,k},\eta_{i2,k},\ldots,\eta_{id,k}\} \in \mathbb{R}^d$ and $\xi_{i,k} = \mathrm{col}\{\xi_{i1,k},\xi_{i2,k},\ldots,\xi_{id,k}\} \in \mathbb{R}^d$ into the updates of $\sigma_{i,k}$ and $y_{i,k}$, respectively. Let $\boldsymbol{\eta}_k = \mathrm{col}\{\eta_{1,k},\eta_{2,k},\ldots,\eta_{N,k}\} \in \mathbb{R}^{Nd}$ and $\boldsymbol{\xi}_k = \mathrm{col}\{\xi_{1,k},\xi_{2,k},\ldots,\xi_{N,k}\} \in \mathbb{R}^{Nd}$. We make the following assumptions regarding the noise.

Assumption 3: The noises $\eta_{i,k},\xi_{i,k} \in \mathbb{R}^d$ are independently drawn by agent i from Laplace distribution, i.e., $\eta_{ij,k} \sim \mathrm{Lap}(\theta_{i,k})$ and $\xi_{ij,k} \sim \mathrm{Lap}(\theta_{i,k})$, and are independent of $\sigma_{i,0}$, $y_{i,0}$. In addition, with each iteration k, the sequence of noise exhibits independence.

For compactness, we introduce the following stacked vectors: $\boldsymbol{x}_k = \mathrm{col}\{x_{1,k},\ldots,x_{N,k}\}$, $\boldsymbol{\sigma}_k = \mathrm{col}\{\sigma_{1,k},\ldots,\sigma_{N,k}\}$, $\boldsymbol{y}_k = \mathrm{col}\{y_{1,k},\ldots,y_{N,k}\}$, $\phi(\boldsymbol{x}_k) = \mathrm{col}\{\phi_1(x_{1,k}),\ldots,\phi_N(x_{N,k})\}$, $\nabla_1 f(\boldsymbol{x}_k,\boldsymbol{\sigma}_k) = \mathrm{col}\{\nabla_1 f_1(x_{1,k},\boldsymbol{\sigma}_k),\ldots,\nabla_1 f_1(x_{N,k},\boldsymbol{\sigma}_k)\}$, $\nabla_2 f(\boldsymbol{x}_k,\boldsymbol{\sigma}_k) = \mathrm{col}\{\nabla_2 f_1(x_{1,k},\boldsymbol{\sigma}_k),\ldots,\nabla_2 f_N(x_{N,k},\boldsymbol{\sigma}_k)\}$. Then, the PP-HB-DAGT algorithm can be rewritten as the following compact form:

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \alpha\left[\nabla_1 f\left(\boldsymbol{x}_k,\boldsymbol{\sigma}_k\right) + \nabla\phi\left(\boldsymbol{x}_k\right)\boldsymbol{y}_k\right]$$

Algorithm 1 Privacy-Preserving Heavy-Ball DAGT

Parameters: step size α; momentum coefficient β; Laplace noises $\eta_{i,k} = \mathrm{col}\{\eta_{i1,k}, \eta_{i2,k}, \ldots, \eta_{id,k}\} \in \mathbb{R}^d$ with $\eta_{ij,k} \sim \mathrm{Lap}(\theta_{i,k})$ and $\xi_{i,k} = \mathrm{col}\{\xi_{i1,k}, \xi_{i2,k}, \ldots, \xi_{id,k}\} \in \mathbb{R}^d$ with $\xi_{ij,k} \sim \mathrm{Lap}(\theta_{i,k})$;

Initialization: Each agent $i \in \mathcal{V}$ initializes with arbitrary states $x_{i,0} \in \mathbb{R}^q$, $\sigma_{i,0} = \phi_i(x_{i,0})$, and $y_{i,0} = \nabla_2 f_i(x_{i,0}, \sigma_{i,0})$.

for $k = 0, 1, \ldots$ **do**

1) Each agent i injects noises $\eta_{i,k}$ and $\xi_{i,k}$ into $\sigma_{i,k}$ and $y_{i,k}$, respectively, and then sends $\sigma_{i,k} + \eta_{i,k}$ and $y_{i,k} + \xi_{i,k}$ to all neighbors $j \in \mathcal{N}_i$;
2) After receiving $\sigma_{j,k} + \eta_{j,k}$ and $y_{j,k} + \xi_{j,k}$ from all neighbors $j \in \mathcal{N}_i$, each agent i updates its states as follows:

$$x_{i,k+1} = x_{i,k} - \alpha\left[\nabla_1 f_i\left(x_{i,k}, \sigma_{i,k}\right) + \nabla\phi_i\left(x_{i,k}\right) y_{i,k}\right] + \beta\left(x_{i,k} - x_{i,k-1}\right) \tag{4a}$$

$$\sigma_{i,k+1} = \sum_{j=1}^{N} a_{ij}\left(\sigma_{j,k} + \eta_{j,k}\right) + \phi_i\left(x_{i,k+1}\right) - \phi_i\left(x_{i,k}\right) - \eta_{i,k} \tag{4b}$$

$$y_{i,k+1} = \sum_{j=1}^{N} a_{ij}\left(y_{j,k} + \xi_{j,k}\right) + \nabla_2 f_i\left(x_{i,k+1}, \sigma_{i,k+1}\right) - \nabla_2 f_i\left(x_{i,k}, \sigma_{i,k}\right) - \xi_{i,k}. \tag{4c}$$

end for

$$+ \beta\left(\boldsymbol{x}_k - \boldsymbol{x}_{k-1}\right) \tag{5a}$$

$$\boldsymbol{\sigma}_{k+1} = \mathcal{A}\left(\boldsymbol{\sigma}_k + \boldsymbol{\eta}_k\right) + \phi\left(\boldsymbol{x}_{k+1}\right) - \phi\left(\boldsymbol{x}_k\right) - \boldsymbol{\eta}_k \tag{5b}$$

$$\boldsymbol{y}_{k+1} = \mathcal{A}\left(\boldsymbol{y}_k + \boldsymbol{\xi}_k\right) + \nabla_2 f\left(\boldsymbol{x}_{k+1}, \boldsymbol{\sigma}_{k+1}\right) - \nabla_2 f\left(\boldsymbol{x}_k, \boldsymbol{\sigma}_k\right) - \boldsymbol{\xi}_k. \tag{5c}$$

Remark 2: It is worth noting that the proposed algorithm (5) adopts a two-stage noise processing mechanism. In the first stage, noise is injected into the transmitted information to achieve privacy preservation. In the second stage, the same noise is systematically subtracted in the dynamic updates of the aggregative estimates and gradient tracking variables. This design is crucial for mitigating the accumulation of perturbation errors caused by repeated noise injection.

To illustrate the necessity of this mechanism, consider a hypothetical variant of the algorithm in which the deduction terms are removed from (5b) and (5c). In this case, the update rules reduce to

$$\boldsymbol{\sigma}_{k+1} = \mathcal{A}\left(\boldsymbol{\sigma}_k + \boldsymbol{\eta}_k\right) + \phi\left(\boldsymbol{x}_{k+1}\right) - \phi\left(\boldsymbol{x}_k\right)$$
$$\boldsymbol{y}_{k+1} = \mathcal{A}\left(\boldsymbol{y}_k + \boldsymbol{\xi}_k\right) + \nabla_2 f\left(\boldsymbol{x}_{k+1}, \boldsymbol{\sigma}_{k+1}\right) - \nabla_2 f\left(\boldsymbol{x}_k, \boldsymbol{\sigma}_k\right).$$

By averaging the updates across all N agents, the global average state $\bar{\sigma}_k$ would evolve as

$$\bar{\sigma}_k = \frac{1}{N}\sum_{i=1}^{N}\sigma_{i,k} = \frac{1}{N}\sum_{i=1}^{N}\phi_i\left(x_{i,k}\right) + \frac{1}{N}\sum_{i=1}^{N}\sum_{l=0}^{k-1}\eta_{i,l}.$$

Similarly, for the gradient tracking variable y, the global average becomes

$$\bar{y}_k = \frac{1}{N}\sum_{i=1}^{N} y_{i,k} = \frac{1}{N}\sum_{i=1}^{N}\nabla_2 f_i\left(x_{i,k}, \sigma_{i,k}\right) + \frac{1}{N}\sum_{i=1}^{N}\sum_{l=0}^{k-1}\xi_{i,l}.$$

Consequently, the perturbation noise accumulates indefinitely, resulting in unbounded variance that precludes the algorithm from converging.

B. Supporting Lemmas

To facilitate the convergence and DP analyses of the proposed Algorithm 1, we present several key lemmas below.

Lemma 1: Under Assumption 1, from the updates in (5b) and (5c), for any $k > 0$, it holds that

$$\bar{\sigma}_k = \frac{1}{N}\sum_{i=1}^{N}\sigma_{i,k} = \frac{1}{N}\sum_{i=1}^{N}\phi_i\left(x_{i,k}\right) \tag{6a}$$

$$\bar{y}_k = \frac{1}{N}\sum_{i=1}^{N} y_{i,k} = \frac{1}{N}\sum_{i=1}^{N}\nabla_2 f_i\left(x_{i,k}, \sigma_{i,k}\right). \tag{6b}$$

Proof: Multiplying both sides of (5b) by $1_N^T/N$ and leveraging the doubly stochastic property of the matrix $\mathcal{A}$, we obtain

$$\bar{\sigma}_{k+1} = \frac{1}{N}\sum_{i=1}^{N}\left(\sigma_{i,k} + \eta_{i,k}\right) + \frac{1}{N}\sum_{i=1}^{N}\phi_i\left(x_{i,k+1}\right) - \frac{1}{N}\sum_{i=1}^{N}\phi_i\left(x_{i,k}\right) - \frac{1}{N}\sum_{i=1}^{N}\eta_{i,k}$$

which further implies that

$$\bar{\sigma}_k - \frac{1}{N}\sum_{i=1}^{N}\phi_i\left(x_{i,k}\right) = \bar{\sigma}_0 - \frac{1}{N}\sum_{i=1}^{N}\phi_i\left(x_{i,0}\right).$$

Given the initialization $\sigma_{i,0} = \phi_i(x_{i,0})$, we conclude $\bar{\sigma}_k = (1/N)\sum_{i=1}^{N}\sigma_{i,k} = (1/N)\sum_{i=1}^{N}\phi(x_{i,k})$. A similar argument using (5c) proves (6b). ∎

Lemma 2 ([13]): Let $f : \mathbb{R}^{Nq} \to \mathbb{R}$ be μ-strongly convex and L-smooth. When $\alpha \in (0, 1/L]$, then for any $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathbb{R}^{Nq}$, we have $\|\boldsymbol{x}_1 - \alpha\nabla f(\boldsymbol{x}_1) - (\boldsymbol{x}_2 - \alpha\nabla f(\boldsymbol{x}_2))\| \leq (1-\mu\alpha)\|\boldsymbol{x}_1 - \boldsymbol{x}_2\|$.

Lemma 3 ([36]): For U, V $\in \mathbb{R}^{p\times q}$, we have the following inequality:

$$\|\mathbf{U} + \mathbf{V}\|^2 \leq 2\|\mathbf{U}\|^2 + 2\|\mathbf{V}\|^2.$$

In addition, for any $\mathbf{U}_1$, $\mathbf{U}_2$, $\mathbf{U}_3 \in \mathbb{R}^{p\times q}$, and constant $\vartheta > 1$, we have

$$\|\mathbf{U}_1 + \mathbf{U}_2 + \mathbf{U}_3\|^2 \leq \vartheta\|\mathbf{U}_1\|^2 + \frac{2\vartheta}{\vartheta - 1}\left[\|\mathbf{U}_2\|^2 + \|\mathbf{U}_3\|^2\right].$$

In particular, by choosing $\vartheta = 3$, we have

$$\|\mathbf{U}_1 + \mathbf{U}_2 + \mathbf{U}_3\|^2 \leq 3\|\mathbf{U}_1\|^2 + 3\|\mathbf{U}_2\|^2 + 3\|\mathbf{U}_3\|^2.$$

Lemma 4 ([37]): Let Assumption 1 hold. Then there exists a constant $\rho \in (0, 1)$, defined by $\rho = \|A - J\|$, such that for any $x \in \mathbb{R}^{Nd}$, we have $\|\mathcal{A}\boldsymbol{x} - \mathcal{J}\boldsymbol{x}\| \leq \rho\|\boldsymbol{x} - \mathcal{J}\boldsymbol{x}\|$.

Lemma 5 ([5]): For any norm $\|\cdot\|$ well-defined on matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$ and $\mathbf{x} \in \mathbb{R}^{n\times p}$, we always have $\|\mathbf{M}\mathbf{x}\| \leq \|\mathbf{M}\|\|\mathbf{x}\|$.

IV. Convergence Analysis

In this section, we analyze the convergence of the proposed algorithm. Specifically, our analysis focuses on four quantities: $\mathbb{E}[\|\boldsymbol{x}_{k+1} - \boldsymbol{x}^*\|^2]$, $\mathbb{E}[\|\boldsymbol{x}_{k+1} - \boldsymbol{x}_k\|^2]$, $\mathbb{E}[\|\boldsymbol{\sigma}_{k+1} - \mathcal{J}\boldsymbol{\sigma}_{k+1}\|^2]$, and $\mathbb{E}[\|\boldsymbol{y}_{k+1} - \mathcal{J}\boldsymbol{y}_{k+1}\|^2]$. These quantities are used to measure the degree of distance from the optimal solution, the state difference, the error of aggregative function, and the error of aggregative gradient, respectively. Based on these metrics, we derive the following proposition, which serves as the theoretical foundation for the convergence results.

Proposition 1: Under Assumptions 1–3, for step size $\alpha \in (0, 1/L_1)$ and momentum coefficient $\beta > 0$, let the sequences $\{\boldsymbol{x}_k\}$, $\{\boldsymbol{y}_k\}$ and $\{\boldsymbol{\sigma}_k\}$ be generated by Algorithm 1. Define $\boldsymbol{v}_k = [\mathbb{E}[\|\boldsymbol{x}_{k+1} - \boldsymbol{x}^*\|^2], \mathbb{E}[\|\boldsymbol{x}_{k+1} - \boldsymbol{x}_k\|^2], \mathbb{E}[\|\boldsymbol{\sigma}_{k+1} - \mathcal{J}\boldsymbol{\sigma}_{k+1}\|^2], \mathbb{E}[\|\boldsymbol{y}_{k+1} - \mathcal{J}\boldsymbol{y}_{k+1}\|^2]]^T$, $\boldsymbol{\Sigma} = b_1\bar{\theta}^2[0, 0, 1+\rho^2, 1+12L_2^2]^T$ and $\Pi(\alpha,\beta) = \begin{bmatrix} 1-\mu\alpha & (a_1/\alpha)\beta^2 & a_2\alpha & a_3\alpha \\ a_4\alpha^2 & 3\beta^2 & a_5\alpha^2 & a_6\alpha^2 \\ \Pi_{31} & a_7\beta^2 & \Pi_{33} & \Pi_{34} \\ \Pi_{41} & a_8\beta^2 & \Pi_{43} & \Pi_{44} \end{bmatrix}$. Then, the following linear system of inequalities holds for all $k \geq 0$:

$$\boldsymbol{v}_{k+1} \leq \Pi(\alpha,\beta)\,\boldsymbol{v}_k + \boldsymbol{\Sigma} \tag{7}$$

where

$$\Pi_{31} = \frac{1}{3}a_4a_7\alpha^2, \quad \Pi_{33} = \rho + \frac{1}{3}a_5a_7\alpha^2, \quad \Pi_{34} = \frac{1}{3}a_6a_7\alpha^2$$
$$\Pi_{41} = \frac{1}{3}a_4a_8\alpha^2, \quad \Pi_{43} = a_9 + \frac{1}{3}a_5a_8\alpha^2$$
$$\Pi_{44} = \rho + \frac{1}{3}a_6a_8\alpha^2$$
$$a_1 = \frac{4}{\mu}, \quad a_2 = \frac{2L_1^2}{\mu}, \quad a_3 = \frac{4\hat{L}_3^2}{\mu}$$
$$a_4 = 6L_1^2\left(1+2\hat{L}_3^2\right)$$
$$a_5 = 12L_1^2, \quad a_6 = 3\hat{L}_3^2, \quad a_7 = \frac{6\hat{L}_3^2}{1-\rho}$$
$$a_8 = \frac{12L_2^2\left(1+3\hat{L}_3^2\right)}{1-\rho}, \quad a_9 = \frac{48L_2^2}{1-\rho}, \quad b_1 = \frac{8Nd}{1-\rho}. \tag{8}$$

Proof:. The proof is structured into the following four steps.

Step 1: Bounding $\mathbb{E}[\|\boldsymbol{x}_{k+1} - \boldsymbol{x}^*\|^2 \mid \mathcal{F}_k]$.

Based on Assumptions 1 and 2, and by applying the update rule in (5a), we obtain

$$\begin{aligned}
&\left\|\boldsymbol{x}_{k+1} - \boldsymbol{x}^*\right\|^2 \\
&= \|\boldsymbol{x}_k - \boldsymbol{x}^* + \beta(\boldsymbol{x}_k - \boldsymbol{x}_{k-1}) \\
&\quad - \alpha\left[\nabla_1 f(\boldsymbol{x}_k, \boldsymbol{\sigma}_k) + \nabla\phi(\boldsymbol{x}_k)\boldsymbol{y}_k\right]\|^2 \\
&\leq \vartheta_1\|\boldsymbol{x}_k - \boldsymbol{x}^* + \alpha\nabla F(\boldsymbol{x}^*) - \alpha(\nabla_1 f(\boldsymbol{x}_k, \boldsymbol{1}_N \otimes \bar{\sigma}_k) \\
&\quad + \nabla\phi(\boldsymbol{x}_k)\left[\boldsymbol{1}_N \otimes \frac{1}{N}\sum_{i=1}^N \nabla_2 f_i\left(x_{i,k}, \bar{\sigma}_k\right)\right]\Big)\Big\|^2 \\
&\quad + \frac{2\vartheta_1\alpha^2}{\vartheta_1 - 1}\Big\|\nabla_1 f(\boldsymbol{x}_k, \boldsymbol{\sigma}_k) + \nabla\phi(\boldsymbol{x}_k)\boldsymbol{1}_N \otimes \bar{y}_k \\
&\quad - \nabla_1 f(\boldsymbol{x}_k, \boldsymbol{1}_N \otimes \bar{\sigma}_k) - \nabla\phi(\boldsymbol{x}_k)\Big[\boldsymbol{1}_N \otimes \\
&\quad \frac{1}{N}\sum_{i=1}^N \nabla_2 f_i\left(x_{i,k}, \bar{\sigma}_k\right)\Big]\Big\|^2 + \frac{4\beta^2\vartheta_1}{\vartheta_1 - 1}\|\boldsymbol{x}_k - \boldsymbol{x}_{k-1}\|^2 \\
&\quad + \frac{4\alpha^2\vartheta_1}{\vartheta_1 - 1}\|\nabla\phi(\boldsymbol{x}_k)\boldsymbol{y}_k - \nabla\phi(\boldsymbol{x}_k)\boldsymbol{1}_N \otimes \bar{y}_k\|^2 \\
&\leq \vartheta_1(1-\mu\alpha)^2\|\boldsymbol{x}_k - \boldsymbol{x}^*\|^2 + \frac{4\vartheta_1\beta^2}{\vartheta_1 - 1}\|\boldsymbol{x}_k - \boldsymbol{x}_{k-1}\|^2 \\
&\quad + \frac{2\vartheta_1\alpha^2 L_1^2}{\vartheta_1 - 1}\|\boldsymbol{\sigma}_k - \boldsymbol{1}_N \otimes \bar{\sigma}_k\|^2 \\
&\quad + \frac{4\vartheta_1\alpha^2}{\vartheta_1 - 1}\|\nabla\phi(\boldsymbol{x}_k)\|^2\|\boldsymbol{y}_k - \boldsymbol{1}_N \otimes \bar{y}_k\|^2 \\
&\leq \vartheta_1(1-\mu\alpha)^2\|\boldsymbol{x}_k - \boldsymbol{x}^*\|^2 + \frac{4\vartheta_1\beta^2}{\vartheta_1 - 1}\|\boldsymbol{x}_k - \boldsymbol{x}_{k-1}\|^2 \\
&\quad + \frac{2\vartheta_1\alpha^2 L_1^2}{\vartheta_1 - 1}\|\boldsymbol{\sigma}_k - \mathcal{J}\boldsymbol{\sigma}_k\|^2 + \frac{4\vartheta_1\alpha^2\hat{L}_3^2}{\vartheta_1 - 1}\|\boldsymbol{y}_k - \mathcal{J}\boldsymbol{y}_k\|^2
\end{aligned} \tag{9}$$

where the first inequality follows from Lemma 3, while the second inequality is the result of combining Assumption 2 with Lemmas 2, 3, and 5. The last two terms of the third inequality are attributed to Assumption 2, $\boldsymbol{1}_N \otimes \bar{\sigma}_k = \mathcal{J}\boldsymbol{\sigma}_k$, and $\boldsymbol{1}_N \otimes \bar{y}_k = \mathcal{J}\boldsymbol{y}_k$.

Taking conditional expectation with respect to $\mathcal{F}_k = \{\eta_0, \ldots, \eta_k; \xi_0, \ldots, \xi_k\}$, choosing $\vartheta_1 = 1/(1-\mu\alpha)$, we have

$$\begin{aligned}
&\mathbb{E}\left[\|\boldsymbol{x}_{k+1} - \boldsymbol{x}^*\|^2 \mid \mathcal{F}_k\right] \\
&\leq (1-\mu\alpha)\,\mathbb{E}\left[\|\boldsymbol{x}_k - \boldsymbol{x}^*\|^2\right] + a_1\frac{\beta^2}{\alpha}\mathbb{E}\left[\|\boldsymbol{x}_k - \boldsymbol{x}_{k-1}\|^2\right] \\
&\quad + a_2\alpha\mathbb{E}\left[\|\boldsymbol{\sigma}_k - \mathcal{J}\boldsymbol{\sigma}_k\|^2\right] + a_3\alpha\mathbb{E}\left[\|\boldsymbol{y}_k - \mathcal{J}\boldsymbol{y}_k\|^2\right]
\end{aligned} \tag{10}$$

where a_1, a_2, a_3 are defined in (8).

Step 2: Bounding $\mathbb{E}[\|\boldsymbol{x}_{k+1} - \boldsymbol{x}_k\|^2 \mid \mathcal{F}_k]$.

Under Assumption 2, and from (5a), we have

$$\begin{aligned}
&\|\boldsymbol{x}_{k+1} - \boldsymbol{x}_k\|^2 \\
&= \left\|\beta(\boldsymbol{x}_k - \boldsymbol{x}_{k-1}) - \alpha\left[\nabla_1 f(\boldsymbol{x}_k, \boldsymbol{\sigma}_k) + \nabla\phi(\boldsymbol{x}_k)\boldsymbol{y}_k\right]\right\|^2 \\
&\leq 3\beta^2\|\boldsymbol{x}_k - \boldsymbol{x}_{k-1}\|^2 + 3\alpha^2\|\nabla_1 f(\boldsymbol{x}_k, \boldsymbol{\sigma}_k) \\
&\quad + \nabla\phi(\boldsymbol{x}_k)\mathcal{J}\boldsymbol{y}_k - \nabla_1 f(\boldsymbol{x}^*, \boldsymbol{1}_N \otimes \sigma^*) \\
&\quad - \nabla\phi(\boldsymbol{x}^*)\left[\boldsymbol{1}_N \otimes \frac{1}{N}\sum_{i=1}^N \nabla_2 f_i\left(x_i^*, \sigma^*\right)\right]\Bigg\|^2 \\
&\quad + 3\alpha^2\|\nabla\phi(\boldsymbol{x}_k)\left(\boldsymbol{y}_k - \mathcal{J}\boldsymbol{y}_k\right)\|^2 \\
&\leq 3\beta^2\|\boldsymbol{x}_k - \boldsymbol{x}_{k-1}\|^2 + 3\alpha^2\hat{L}_3^2\|\boldsymbol{y}_k - \mathcal{J}\boldsymbol{y}_k\|^2 \\
&\quad + 3\alpha^2 L_1^2(\|\boldsymbol{x}_k - \boldsymbol{x}^*\| + \|\boldsymbol{\sigma}_k - \boldsymbol{1}_N \otimes \sigma^*\|)^2
\end{aligned} \tag{11}$$

where the first inequality follows from the optimality condition $\nabla F(\boldsymbol{x}^*) = \nabla_1 f(\boldsymbol{x}^*, \boldsymbol{1}_N \otimes \sigma^*) + \nabla\phi(\boldsymbol{x}^*)\left[\boldsymbol{1}_N \otimes (1/N)\sum_{i=1}^N \nabla_2 f_i(x_i^*, \sigma^*)\right] = \boldsymbol{0}$ and Lemma 3. Meanwhile, note that $\|\boldsymbol{\sigma}_k - \boldsymbol{1}_N \otimes \sigma^*\|^2 \leq 2(\|\boldsymbol{\sigma}_k - \mathcal{J}\boldsymbol{\sigma}_k\|^2 + \|\mathcal{J}\boldsymbol{\sigma}_k - \boldsymbol{1}_N \otimes \sigma^*\|^2)$ hold. Furthermore, based on Lemma 1, we obtain

$$\begin{aligned}
\|\mathcal{J}\boldsymbol{\sigma}_k - \boldsymbol{1}_N \otimes \sigma^*\|^2 &= \|\boldsymbol{1}_N \otimes (\bar{\sigma}_k - \sigma^*)\|^2 \\
&= N\left\|\frac{1}{N}\sum_{i=1}^N \left(\phi_i\left(x_{i,k}\right) - \phi_i\left(x_i^*\right)\right)\right\|^2 \\
&\leq \frac{1}{N}\left(\sum_{i=1}^N \left\|\phi_i\left(x_{i,k}\right) - \phi_i\left(x_i^*\right)\right\|\right)^2 \\
&\leq \frac{1}{N}\left(\sum_{i=1}^N L_{3,i}\left\|x_{i,k} - x_i^*\right\|\right)^2 \\
&\leq \sum_{i=1}^N L_{3,i}^2\left\|x_{i,k} - x_i^*\right\|^2 \\
&\leq \hat{L}_3^2\|\boldsymbol{x}_k - \boldsymbol{x}^*\|^2
\end{aligned} \tag{12}$$

where Lemma 3 is used in the first inequality, Assumption 2 is used in the second inequality, and for any nonnegative scalar a_i's, $\left(\sum_{i=1}^N a_i\right)^2 \leq N\sum_{i=1}^N a_i^2$ is used in the third inequality.

Substituting (12) into (11), we derive

$$\left\|\boldsymbol{x}_{k+1} - \boldsymbol{x}_k\right\|^2$$

$$\leq 6\alpha^2 L_1^2\left(1+2\hat{L}_3^2\right)\|\boldsymbol{x}_k-\boldsymbol{x}^*\|^2+3\beta^2\|\boldsymbol{x}_k-\boldsymbol{x}_{k-1}\|^2 + 12\alpha^2 L_1^2\|\boldsymbol{\sigma}_k-\mathcal{J}\boldsymbol{\sigma}_k\|^2+3\alpha^2\hat{L}_3^2\|\boldsymbol{y}_k-\mathcal{J}\boldsymbol{y}_k\|^2. \tag{13}$$

Then, taking conditional expectation with respect to $\mathcal{F}_k = \{\eta_0,\ldots,\eta_k;\xi_0,\ldots,\xi_k\}$, we get

$$\begin{aligned}&\mathbb{E}\left[\|\boldsymbol{x}_{k+1}-\boldsymbol{x}_k\|^2\mid\mathcal{F}_k\right]\\&\leq a_4\alpha^2\mathbb{E}\left[\|\boldsymbol{x}_k-\boldsymbol{x}^*\|^2\right]+3\beta^2\mathbb{E}\left[\|\boldsymbol{x}_k-\boldsymbol{x}_{k-1}\|^2\right]\\&\quad+a_5\alpha^2\mathbb{E}\left[\|\boldsymbol{\sigma}_k-\mathcal{J}\boldsymbol{\sigma}_k\|^2\right]+a_6\alpha^2\mathbb{E}\left[\|\boldsymbol{y}_k-\mathcal{J}\boldsymbol{y}_k\|^2\right]\end{aligned}\tag{14}$$

where a_4, a_5, a_6 are defined in (8).

Step 3: Bounding $\mathbb{E}[\|\boldsymbol{\sigma}_{k+1}-\mathcal{J}\boldsymbol{\sigma}_{k+1}\|^2\mid\mathcal{F}_k]$.

Under Assumptions 1 and 2, and based on (5b), we obtain

$$\begin{aligned}&\|\boldsymbol{\sigma}_{k+1}-\mathcal{J}\boldsymbol{\sigma}_{k+1}\|^2\\&=\|\mathcal{A}\left(\boldsymbol{\sigma}_k+\boldsymbol{\eta}_k\right)+\phi\left(\boldsymbol{x}_{k+1}\right)-\phi\left(\boldsymbol{x}_k\right)-\boldsymbol{\eta}_k\\&\quad-\mathcal{J}\mathcal{A}\left(\boldsymbol{\sigma}_k+\boldsymbol{\eta}_k\right)-\mathcal{J}\left(\phi\left(\boldsymbol{x}_{k+1}\right)-\phi\left(\boldsymbol{x}_k\right)\right)+\mathcal{J}\boldsymbol{\eta}_k\|^2\\&\leq\vartheta_2\rho^2\|\boldsymbol{\sigma}_k-\mathcal{J}\boldsymbol{\sigma}_k\|^2+\frac{4\vartheta_2\left(1+\rho^2\right)}{\vartheta_2-1}\|\boldsymbol{\eta}_k-\mathcal{J}\boldsymbol{\eta}_k\|^2\\&\quad+\frac{2\vartheta_2}{\vartheta_2-1}\|I_{Nd}-\mathcal{J}\|^2\|\phi\left(\boldsymbol{x}_{k+1}\right)-\phi\left(\boldsymbol{x}_k\right)\|^2\\&\leq\vartheta_2\rho^2\|\boldsymbol{\sigma}_k-\mathcal{J}\boldsymbol{\sigma}_k\|^2+\frac{4\vartheta_2\left(1+\rho^2\right)}{\vartheta_2-1}\|\boldsymbol{\eta}_k-\mathcal{J}\boldsymbol{\eta}_k\|^2\\&\quad+\frac{2\vartheta_2\hat{L}_3^2}{\vartheta_2-1}\|I_{Nd}-\mathcal{J}\|^2\|\boldsymbol{x}_{k+1}-\boldsymbol{x}_k\|^2\end{aligned}\tag{15}$$

where the first inequality follows from Lemmas 3 to 5.

By substituting (13) into (15), choosing $\vartheta_2 = 1/\rho$, considering $\|I_{Nd}-\mathcal{J}\|^2 = 1$ and then taking conditional expectation with respect to $\mathcal{F}_k = \{\eta_0,\ldots,\eta_k;\xi_0,\ldots,\xi_k\}$, we obtain

$$\begin{aligned}&\mathbb{E}\left[\|\boldsymbol{\sigma}_{k+1}-\mathcal{J}\boldsymbol{\sigma}_{k+1}\|^2\mid\mathcal{F}_k\right]\\&\leq\frac{1}{3}a_4a_7\alpha^2\mathbb{E}\left[\|\boldsymbol{x}_k-\boldsymbol{x}^*\|^2\right]+a_7\beta^2\mathbb{E}\left[\|\boldsymbol{x}_k-\boldsymbol{x}_{k-1}\|^2\right]\\&\quad+\left(\rho+\frac{1}{3}a_5a_7\alpha^2\right)\mathbb{E}\left[\|\boldsymbol{\sigma}_k-\mathcal{J}\boldsymbol{\sigma}_k\|^2\right]\\&\quad+\frac{1}{3}a_6a_7\alpha^2\mathbb{E}\left[\|\boldsymbol{y}_k-\mathcal{J}\boldsymbol{y}_k\|^2\right]+b_1\bar{\theta}^2\left(1+\rho^2\right)\end{aligned}\tag{16}$$

where a_7 is defined in (8) and $\bar{\theta} = *max_i\theta_i$.

Step 4: Bounding $\mathbb{E}[\|\boldsymbol{y}_{k+1}-\mathcal{J}\boldsymbol{y}_{k+1}\|^2\mid\mathcal{F}_k]$.

Under Assumptions 1 and 2, and based on (5c), we get

$$\begin{aligned}&\|\boldsymbol{y}_{k+1}-\mathcal{J}\boldsymbol{y}_{k+1}\|^2\\&=\big\|\mathcal{A}(\boldsymbol{y}_k+\boldsymbol{\xi}_k)-\mathcal{J}\mathcal{A}(\boldsymbol{y}_k+\boldsymbol{\xi}_k)-\boldsymbol{\xi}_k\\&\quad+\nabla_2 f\left(\boldsymbol{x}_{k+1},\boldsymbol{\sigma}_{k+1}\right)-\nabla_2 f\left(\boldsymbol{x}_k,\boldsymbol{\sigma}_k\right)+\mathcal{J}\boldsymbol{\xi}_k\\&\quad-\mathcal{J}\left[\nabla_2 f\left(\boldsymbol{x}_{k+1},\boldsymbol{\sigma}_{k+1}\right)-\nabla_2 f\left(\boldsymbol{x}_k,\boldsymbol{\sigma}_k\right)\right]\big\|^2\\&\leq\frac{2\vartheta_3}{\vartheta_3-1}\|I_{Nd}-\mathcal{J}\|^2\|\nabla_2 f\left(\boldsymbol{x}_{k+1},\boldsymbol{\sigma}_{k+1}\right)\\&\quad-\nabla_2 f\left(\boldsymbol{x}_k,\boldsymbol{\sigma}_k\right)\|^2+\vartheta_3\rho^2\|\boldsymbol{y}_k-\mathcal{J}\boldsymbol{y}_k\|^2\\&\quad+\frac{2\vartheta_3(1+\rho^2)}{\vartheta_3-1}\left\|\boldsymbol{\xi}_k-\mathcal{J}\boldsymbol{\xi}_k\right\|^2\end{aligned}\tag{17}$$

where the inequality is derived by applying Lemmas 3–5.

In addition, we have $\|\nabla_2 f(\boldsymbol{x}_{k+1},\boldsymbol{\sigma}_{k+1})-\nabla_2 f(\boldsymbol{x}_k,\boldsymbol{\sigma}_k)\|^2 \leq 2L_2^2(\|\boldsymbol{x}_{k+1}-\boldsymbol{x}_k\|^2+\|\boldsymbol{\sigma}_{k+1}-\boldsymbol{\sigma}_k\|^2)$, and

$$\begin{aligned}&\|\boldsymbol{\sigma}_{k+1}-\boldsymbol{\sigma}_k\|^2\\&=\|\mathcal{A}\left(\boldsymbol{\sigma}_k+\boldsymbol{\eta}_k\right)-\boldsymbol{\sigma}_k+\phi\left(\boldsymbol{x}_{k+1}\right)-\phi\left(\boldsymbol{x}_k\right)-\boldsymbol{\eta}_k\|^2\\&=\|(\mathcal{A}-I\otimes I_d)(\boldsymbol{\sigma}_k-\mathcal{J}\boldsymbol{\sigma}_k)\\&\quad+\mathcal{A}\boldsymbol{\eta}_k+\phi\left(\boldsymbol{x}_{k+1}\right)-\phi\left(\boldsymbol{x}_k\right)-\boldsymbol{\eta}_k\|^2\\&\leq 3\|A-I\|^2\|\boldsymbol{\sigma}_k-\mathcal{J}\boldsymbol{\sigma}_k\|^2+3\hat{L}_3^2\|\boldsymbol{x}_{k+1}-\boldsymbol{x}_k\|^2\\&\quad+3\|A-I\|^2\|\boldsymbol{\eta}_k\|^2\end{aligned}\tag{18}$$

where the fact that $\mathcal{A}\boldsymbol{\sigma}_k-\boldsymbol{\sigma}_k = (\mathcal{A}-I\otimes I_d)(\boldsymbol{\sigma}_k-\mathcal{J}\boldsymbol{\sigma}_k)$ is used in the second equality. Then, we get

$$\begin{aligned}&\|\nabla_2 f\left(\boldsymbol{x}_{k+1},\boldsymbol{\sigma}_{k+1}\right)-\nabla_2 f\left(\boldsymbol{x}_k,\boldsymbol{\sigma}_k\right)\|^2\\&\leq 2L_2^2\left(1+3\hat{L}_3^2\right)\|\boldsymbol{x}_{k+1}-\boldsymbol{x}_k\|^2+6L_2^2\|A-I\|^2\|\boldsymbol{\sigma}_k-\mathcal{J}\boldsymbol{\sigma}_k\|^2\\&\quad+6L_2^2\|A-I\|^2\|\boldsymbol{\eta}_k\|^2.\end{aligned}\tag{19}$$

By selecting $\vartheta_3 = 1/\rho$, substituting (13) and (19) into (17), and using $\|A-I\|^2 \leq 4$, we take conditional expectation with respect to $\mathcal{F}_k = \{\eta_0,\ldots,\eta_k;\xi_0,\ldots,\xi_k\}$, and obtain

$$\begin{aligned}&\mathbb{E}\left[\|\boldsymbol{y}_{k+1}-\mathcal{J}\boldsymbol{y}_{k+1}\|^2\mid\mathcal{F}_k\right]\\&\leq\frac{1}{3}a_4a_8\alpha^2\mathbb{E}\left[\|\boldsymbol{x}_k-\boldsymbol{x}^*\|^2\right]+a_8\beta^2\mathbb{E}\left[\|\boldsymbol{x}_k-\boldsymbol{x}_{k-1}\|^2\right]\\&\quad+\left(a_9+\frac{1}{3}a_5a_8\alpha^2\right)\mathbb{E}\left[\|\boldsymbol{\sigma}_k-\mathcal{J}\boldsymbol{\sigma}_k\|^2\right]\\&\quad+\left(\rho+\frac{1}{3}a_6a_8\alpha^2\right)\mathbb{E}\left[\|\boldsymbol{y}_k-\mathcal{J}\boldsymbol{y}_k\|^2\right]\\&\quad+b_1\bar{\theta}^2\left(1+12L_2^2\right)\end{aligned}\tag{20}$$

where a_8, a_9 are defined in (8).

Step 5: Taking full expectations of (10), (14), (16), and (20), and combining them accordingly, we can obtain the inequality (1) directly. ■

The following theorem shows the convergence properties of Algorithm 1.

Theorem 1: Suppose Assumptions 1–3 hold. If the step size α and the momentum coefficient β satisfy $0 < \alpha < \min\{(1/L_1),\varpi_5,\varpi_6,\varpi_7,\varpi_8,\varpi_9\}$ and $0 < \beta < \min\{\varpi_1,\varpi_2,\varpi_3,\varpi_4\}$, where $\varpi_1,\cdots,\varpi_9$ are defined in (22) and (24), then $\sup_{l\geq k}\mathbb{E}[\|x_l-x^*\|^2]$ converges to $\limsup_{k\to\infty}\mathbb{E}[\|x_k-x^*\|^2]$ with a linear rate $\mathcal{O}(\rho(\Pi(\alpha,\beta))^k)$, where $\rho(\Pi(\alpha,\beta)) < 1$. Furthermore, the asymptotic bound on the expected optimality gap is given by

$$\limsup_{k\to\infty}\mathbb{E}\left[\|x_k-x^*\|^2\right]\leq\pi_1 \tag{21}$$

where π_1 is the first element of the vector $(I-\Pi(\alpha,\beta))^{-1}\Sigma$. The matrix $\Pi(\alpha,\beta)$ and Σ are defined in Proposition 1.

Proof: By induction, the recursion in (1) yields $\upsilon_{k+1} \leq \Pi(\alpha,\beta)^k\upsilon_0 + \sum_{l=0}^{k-1}\Pi(\alpha,\beta)^l\Sigma$. Thus, to establish linear convergence of υ_k, it suffices to show that the spectral radius satisfies $\rho(\Pi(\alpha,\beta)) < 1$. According to the Perron–Frobenius result in [33, Corollary 8.1.29], this condition holds if there exists a strictly positive vector $\zeta = (\zeta_1,\zeta_2,\zeta_3,\zeta_4)^T \in \mathbb{R}^4$ such that $\Pi(\alpha,\beta)\zeta < \zeta$.

To this end, select $\zeta > 0$ satisfying $\zeta_1 > (a_2\zeta_3+a_3\zeta_4/\mu), \zeta_2 > 0, \zeta_3 > 0, \zeta_4 > (a_9\zeta_3/1-\rho)$.

Define

$$\varpi_5 = \sqrt{\frac{\zeta_2}{a_4\zeta_1+a_5\zeta_3+a_6\zeta_4}},\quad \varpi_6 = (1-\rho)\sqrt{\frac{3}{2a_5a_6}}$$

$$\varpi_7 = \sqrt{\frac{3\left(1-\rho\right)\zeta_3}{\left(a_4\zeta_1+a_5\zeta_3+a_6\zeta_4\right)a_7}}$$

$$\varpi_8 = \sqrt{\frac{3\left[(1-\rho)\zeta_4 - a_9\zeta_3\right]}{(a_4\zeta_1 + a_5\zeta_3 + a_6\zeta_4)a_8}}, \quad \varpi_9 = \sqrt{\frac{3(1-\rho)}{a_6 a_8}}. \tag{22}$$

Substituting ζ into $\Pi(\alpha,\beta)\zeta < \zeta$ yields the componentwise inequalities

$$a_1\frac{\beta^2}{\alpha}\zeta_2 < \mu\alpha\zeta_1 - a_2\alpha\zeta_3 - \alpha a_3\zeta_4 \tag{23a}$$
$$3\beta^2\zeta_2 < \zeta_2 - \alpha^2\left[a_4\zeta_1 + a_5\zeta_3 + a_6\zeta_4\right] \tag{23b}$$
$$a_7\beta^2\zeta_2 < \left(1-\rho-\frac{1}{3}a_5a_7\alpha^2\right)\zeta_3 - \frac{1}{3}a_4a_7\alpha^2\zeta_1 - \frac{1}{3}a_6a_7\alpha^2\zeta_4 \tag{23c}$$
$$a_8\beta^2\zeta_2 < \left(1-\rho-\frac{1}{3}a_6a_8\alpha^2\right)\zeta_4 - \frac{1}{3}a_4a_8\alpha^2\zeta_1 - \left(a_9 + \frac{1}{3}a_5a_8\alpha^2\right)\zeta_3. \tag{23d}$$

If the step size α satisfies $0 < \alpha < \min\{(1/L_1), \varpi_5, \varpi_6, \varpi_7, \varpi_8, \varpi_9\}$, then all right-hand sides of (23) are strictly positive. Rearranging each inequality in (23) and taking square roots yields

$$\beta < \alpha\sqrt{\frac{\mu\zeta_1 - a_2\zeta_3 - a_3\zeta_4}{a_1\zeta_2}} \triangleq \varpi_1 \tag{24a}$$
$$\beta < \sqrt{\frac{\zeta_2 - \alpha^2(a_4\zeta_1 + a_5\zeta_3 + a_6\zeta_4)}{3\zeta_2}} \triangleq \varpi_2 \tag{24b}$$
$$\beta < \sqrt{\frac{\left[3(1-\rho) - a_5a_7\alpha^2\right]\zeta_3 - a_4a_7\alpha^2\zeta_1 - a_6a_7\alpha^2\zeta_4}{3a_7\zeta_2}} \triangleq \varpi_3 \tag{24c}$$
$$\beta < \sqrt{\frac{\left[3(1-\rho) - a_6a_8\alpha^2\right]\zeta_4 - a_4a_8\alpha^2\zeta_1 - \left(3a_9 + a_5a_8\alpha^2\right)\zeta_3}{3a_8\zeta_2}} \triangleq \varpi_4. \tag{24d}$$

Therefore, if $0 < \beta < \min\{\varpi_1, \varpi_2, \varpi_3, \varpi_4\}$, all the inequalities in (24) hold simultaneously, and hence $\Pi(\alpha,\beta)\zeta < \zeta$. Since $\zeta > 0$ and $\Pi(\alpha,\beta)$ is nonnegative, if follows from [37, Corollary 8.1.29] that $\rho(\Pi(\alpha,\beta)) < 1$. Consequently, $\Pi(\alpha,\beta)^k$ decays to zero at a linear rate, and the Neumann series $\sum_{l=0}^{\infty}\Pi(\alpha,\beta)^l = (I-\Pi(\alpha,\beta))^{-1}$ converges. Combining this with the affine recursion implies that $\lim_{k\to\infty} \upsilon_k \le (I-\Pi(\alpha,\beta))^{-1}\Sigma$. Since the first component of υ_k corresponds to $\mathbb{E}\|x_k - x^*\|^2$, we obtain the asymptotic bound $\lim_{k\to\infty}\sup \mathbb{E}\|x_k - x^*\|^2 \le [(I-\Pi(\alpha,\beta))^{-1}\Sigma]_1 \triangleq \pi_1$, which completes the proof. ■

To simplify the asymptotic upper bound expression in (21), we define $c_1 = -3a_3(1+12L_2^2)$, $c_2 = (a_3a_5 - a_2a_6)a_7(1+12L_2^2)\alpha^2$, $c_3 = -3a_2(1+9\beta^2)(1+\rho^2)$, $c_4 = [(a_2a_6 - a_3a_5)a_8\alpha^2 - 3a_3a_9](1+\rho^2)$. Then, the first element of the vector $(I-\Pi(\alpha,\beta))^{-1}\Sigma$ is given by

$$\left[(I-\Pi(\alpha,\beta))^{-1}\Sigma\right]_1 = \frac{\varphi_1 T + \varphi_2}{\kappa_1 T^2 + \kappa_2 T + \kappa_3}e_1 \tag{25}$$

where $T = 1-\rho$, $e_1 = b_1\bar{\theta}^2$, $\varphi_1 = c_1 + c_2$, $\varphi_2 = c_3 + c_4$, $\kappa_1 = 3(3\beta^2\mu + a_1a_4\beta^2 + \mu)$, $\kappa_2 = [(a_2a_7 + a_3a_8)a_4 + (a_5a_7 + a_6a_8)]a_7a_9\alpha^2$, and $\kappa_3 = (a_3a_4 + a_6\mu)\alpha^2$.

Remark 3: According to Theorem 1, if the noise level is set to decay over time, we have $\boldsymbol{\Sigma}_k = b_1\bar{\theta}_k^2[0, 0, 1+\rho^2, 1+12L_2^2]^T$, where $\sum_{k=0}^{\infty}\bar{\theta}_k^2 < \infty$. Then, based on [38], it can be shown that the proposed Algorithm 1 converges to the optimal solution x^*. However, reducing the noise level degrades privacy protection. To ensure strong privacy guarantees, this article adopts a constant noise level throughout the process.

V. DP Analysis

In this section, we prove that the proposed algorithm ensures ϵ-DP. Let $\mathcal{H}_k$ denote the set of messages exchanged among agents at iteration k, defined as $\mathcal{H}_k = \{\sigma_{i,k} + \eta_{i,k}, y_{i,k} + \xi_{i,k} | \forall i \in \mathcal{V}\}$.

To establish DP, we impose the following assumption.

Assumption 4: For any $x, y \in \mathbb{R}^n$, assume that $\phi_{i_0}^{(1)}(x) - \phi_{i_0}^{(1)}(y) = \phi_{i_0}^{(2)}(x) - \phi_{i_0}^{(2)}(y)$ and $\nabla_2 f_{i_0}^{(1)}(x, \sigma(\boldsymbol{x})) - \nabla_2 f_{i_0}^{(1)}(y, \sigma(\boldsymbol{y})) = \nabla_2 f_{i_0}^{(2)}(x, \sigma(\boldsymbol{x})) - \nabla_2 f_{i_0}^{(2)}(y, \sigma(\boldsymbol{y}))$.

Under Assumption 4, $\phi_{i_0}^{(1)}$ and $\phi_{i_0}^{(2)}$ share the identical smooth constant L_{3,i_0}, and $\nabla_2 f_{i_0}^{(1)}$ and $\nabla_2 f_{i_0}^{(2)}$ share the identical smooth constant L_{2,i_0}.

We now present the following theorem.

Theorem 2: Suppose that Assumptions 1–3 hold. For any finite number of iterations K, Algorithm 1 guarantees ϵ_i-DP for a given $\epsilon_i > 0$ of each agent i's decision variable if the noise parameter in (4) satisfies $\theta_i > K(d)^{1/2}\delta(L_{3,i} + L_{2,i}(1 + (d)^{1/2}L_{3,i})/\epsilon_i)$.

Proof: According to Algorithm 1, given the communication graph $\mathcal{G}$, the initial conditions $\{\boldsymbol{x_0}, \boldsymbol{\sigma_0}, \boldsymbol{y_0}\}$, and the variable set Ψ, the observation sequence $z = \{\mathcal{H}_k\}_{k=0}^K$ is uniquely determined by the noise sequences $\boldsymbol{\eta} = \{\boldsymbol{\eta}_k\}_{k=0}^K$ and $\boldsymbol{\xi} = \{\boldsymbol{\xi}_k\}_{k=0}^K$, i.e., $z = \mathcal{M}_{\mathcal{F}}(\boldsymbol{\eta}, \boldsymbol{\xi})$, where $\mathcal{F} = \{\boldsymbol{x_0}, \boldsymbol{\sigma_0}, \boldsymbol{y_0}, \mathcal{G}, \Psi\}$. From Definition 2, ϵ-DP holds if, for any adjacent variable sets $\Psi^{(1)}$ and $\Psi^{(2)}$, and for any measurable set $\mathcal{H} \subseteq \mathrm{Range}(\mathcal{M})$, the following inequality holds:

$$P\left(\{\boldsymbol{\eta}, \boldsymbol{\xi}\} \in \Omega | \mathcal{M}_{\mathcal{F}^{(1)}}(\boldsymbol{\eta}, \boldsymbol{\xi}) \in \mathcal{H}\right) \le e^{\epsilon}P\left(\{\boldsymbol{\eta}, \boldsymbol{\xi}\} \in \Omega | \mathcal{M}_{\mathcal{F}^{(2)}}(\boldsymbol{\eta}, \boldsymbol{\xi}) \in \mathcal{H}\right) \tag{26}$$

where $\mathcal{F}^{(l)} = \{\boldsymbol{x_0}, \boldsymbol{\sigma_0}, \boldsymbol{y_0}, \mathcal{G}, \Psi^{(l)}\}$, $l = 1, 2$, and Ω denotes the sample space.

Denote $\boldsymbol{\eta}^{(l)} = \{\boldsymbol{\eta}_k^{(l)}\}_{k=0}^K$ and $\boldsymbol{\xi}^{(l)} = \{\boldsymbol{\xi}_k^{(l)}\}_{k=0}^K$ for $l = 1, 2$. Suppose that the observations are the same under both the inputs, i.e., $\mathcal{M}_{\mathcal{F}^{(1)}}(\boldsymbol{\eta}^{(1)}, \boldsymbol{\xi}^{(1)}) = \mathcal{M}_{\mathcal{F}^{(2)}}(\boldsymbol{\eta}^{(2)}, \boldsymbol{\xi}^{(2)})$, which implies $\phi_i^{(1)}(x_{i,k}^{(1)}) + \eta_{i,k}^{(1)} = \phi_i^{(2)}(x_{i,k}^{(2)}) + \eta_{i,k}^{(2)}$, and $\nabla_2 f_i^{(1)}(x_{i,k}^{(1)}, \sigma_{i,k}^{(1)}) + \xi_{i,k}^{(1)} = \nabla_2 f_i^{(2)}(x_{i,k}^{(2)}, \sigma_{i,k}^{(2)}) + \xi_{i,k}^{(2)}$ for all $i \in \mathcal{V}$. Given that the two inputs differ only at agent i_0, and the initialization satisfies $x_{i,0}^{(1)} = x_{i,0}^{(2)}$, $y_i^{(1)} = y_i^{(2)}$ and $\sigma_i^{(1)} = \sigma_i^{(2)}$ for all $i \neq i_0$, it follows that

$$\eta_{i,k}^{(1)} = \eta_{i,k}^{(2)}, \quad \xi_{i,k}^{(1)} = \xi_{i,k}^{(2)} \quad \forall i \neq i_0. \tag{27}$$

Moreover, for the differing agent i_0, one has

$$\Delta\eta_{i_0,k} = -\Delta\sigma_{i_0,k}, \ \Delta\xi_{i_0,k} = -\Delta y_{i_0,k} \tag{28}$$

where $\Delta\eta_{i_0,k} = \eta_{i_0,k}^{(1)} - \eta_{i_0,k}^{(2)}$, $\Delta\sigma_{i_0,k} = \sigma_{i_0,k}^{(1)} - \sigma_{i_0,k}^{(2)}$, $\Delta\xi_{i_0,k} = \xi_{i_0,k}^{(1)} - \xi_{i_0,k}^{(2)}$, and $\Delta y_{i_0,k} = y_{i_0,k}^{(1)} - y_{i_0,k}^{(2)}$.

According to (4), we further have

$$\Delta\sigma_{i_0,k+1} = \Delta\phi_{i_0,k+1} - \Delta\phi_{i_0,k} - \Delta\eta_{i_0,k}$$
$$\Delta y_{i_0,k+1} = \Delta\tau_{i_0,k+1} \tag{29}$$

where $\tau_{i_0,k+1}^{(l)} = \nabla_{2,i_0}^{(l)} f(\boldsymbol{x}_{k+1}, \boldsymbol{\sigma}_{k+1}) - \nabla_{2,i_0}^{(l)} f(\boldsymbol{x}_k, \boldsymbol{\sigma}_k) - \xi_{i_0,k}^{(l)}$, $l = 1, 2$, $\Delta\phi_{i_0,k} = \phi_{i_0,k}^{(1)} - \phi_{i_0,k}^{(2)}$, and $\Delta\tau_{i_0,k+1} = \tau_{i_0,k+1}^{(1)} - \tau_{i_0,k+1}^{(2)}$.

According to (27)–(29), we know that for each $\{\boldsymbol{\eta}^{(1)},\boldsymbol{\xi}^{(1)}\}$, there exists a unique $\{\boldsymbol{\eta}^{(2)},\boldsymbol{\xi}^{(2)}\}$ such that the sequences of observations generated by $\{\boldsymbol{\eta}^{(1)},\boldsymbol{\xi}^{(1)}\}$ and $\{\boldsymbol{\eta}^{(2)},\boldsymbol{\xi}^{(2)}\}$ are the same. Let $\mathcal{B}(\cdot)$ be a mapping that satisfies $\mathcal{B}(\{\boldsymbol{\eta}^{(1)},\boldsymbol{\xi}^{(1)}\}) = \{\boldsymbol{\eta}^{(2)},\boldsymbol{\xi}^{(2)}\}$ if and only if the resulting sequences of observations are the same. It is clear that $\mathcal{B}(\cdot)$ is bijective. Let $\Upsilon^{(l)} = \{\{\boldsymbol{\eta},\boldsymbol{\xi}\}\in\Omega|\mathcal{M}_{\mathcal{F}^{(l)}}(\boldsymbol{\eta},\boldsymbol{\xi})\in\mathcal{H}\}$, then following the analysis in [28], we obtain

$$\begin{aligned}&\frac{P\left(\{\boldsymbol{\eta},\boldsymbol{\xi}\}\in\Omega|\mathcal{M}_{\mathcal{F}^{(1)}}\left(\boldsymbol{\eta},\boldsymbol{\xi}\right)\in\mathcal{H}\right)}{P\left(\{\boldsymbol{\eta},\boldsymbol{\xi}\}\in\Omega|\mathcal{M}_{\mathcal{F}^{(2)}}\left(\boldsymbol{\eta},\boldsymbol{\xi}\right)\in\mathcal{H}\right)}\\&=\frac{\int\int_{\Upsilon^{(1)}} f_{\eta\xi}\left(\boldsymbol{\eta}^{(1)},\boldsymbol{\xi}^{(1)}\right)d\boldsymbol{\eta}^{(1)}d\boldsymbol{\xi}^{(1)}}{\int\int_{\Upsilon^{(2)}} f_{\eta\xi}\left(\boldsymbol{\eta}^{(2)},\boldsymbol{\xi}^{(2)}\right)d\boldsymbol{\eta}^{(2)}d\boldsymbol{\xi}^{(2)}}\\&=\frac{\int\int_{\Upsilon^{(1)}} f_{\eta\xi}\left(\boldsymbol{\eta}^{(1)},\boldsymbol{\xi}^{(1)}\right)d\boldsymbol{\eta}^{(1)}d\boldsymbol{\xi}^{(1)}}{\int\int_{\Upsilon^{(1)}} f_{\eta\xi}\left(\mathcal{B}\left(\{\boldsymbol{\eta}^{(1)},\boldsymbol{\xi}^{(1)}\}\right)\right)d\boldsymbol{\eta}^{(1)}d\boldsymbol{\xi}^{(1)}}\end{aligned}\tag{30}$$

in which

$$\begin{aligned}&\frac{f_{\eta\xi}(\boldsymbol{\eta}^{(1)},\boldsymbol{\xi}^{(1)})}{f_{\eta\xi}\left(\mathcal{B}\left(\{\boldsymbol{\eta}^{(1)},\boldsymbol{\xi}^{(1)}\}\right)\right)}\\&=\prod_{i=1}^{N}\prod_{k=0}^{K}\frac{f_L(\eta_{i,k}^{(1)}|\theta_i)f_L(\xi_{i,k}^{(1)}|\theta_i)}{f_L(\eta_{i,k}^{(2)}|\theta_i)f_L(\xi_{i,k}^{(2)}|\theta_i)}\\&=\prod_{j=1}^{q}\prod_{k=0}^{K}\frac{f_L(\eta_{i_0j,k}^{(1)}|\theta_{i_0})f_L(\xi_{i_0j,k}^{(1)}|\theta_{i_0})}{f_L(\eta_{i_0j,k}^{(2)}|\theta_{i_0})f_L(\xi_{i_0j,k}^{(2)}|\theta_{i_0})}\\&=\prod_{k=0}^{K}\exp\left(\sum_{j=1}^{q}\left(\frac{|\eta_{i_0j,k}^{(1)}-\eta_{i_0j,k}^{(2)}|}{\theta_{i_0}}+\frac{|\xi_{i_0j,k}^{(1)}-\xi_{i_0j,k}^{(2)}|}{\theta_{i_0}}\right)\right)\\&=\prod_{k=0}^{K}\exp\left(\frac{\|\Delta\eta_{i_0,k}\|_1+\|\Delta\xi_{i_0,k}\|_1}{\theta_{i_0}}\right).\end{aligned}\tag{31}$$

According to (28), (29), and Assumptions 2, 3, it can be deduced that

$$\begin{aligned}\|\Delta\eta_{i_0,k}\|_1 &= \|\Delta\sigma_{i_0,k}\|_1\\&\le \sqrt{d}\|\phi_{i_0}^{(1)}\left(x_{i_0,k}^{(1)}\right)-\phi_{i_0}^{(1)}\left(x_{i_0,k-1}^{(1)}\right)-\eta_{i_0,k-1}^{(1)}\\&\quad-\phi_{i_0}^{(2)}\left(x_{i_0,k}^{(2)}\right)+\phi_{i_0}^{(2)}\left(x_{i_0,k-1}^{(2)}\right)+\eta_{i_0,k-1}^{(2)}\|\\&= \sqrt{d}\|\phi_{i_0}^{(1)}\left(x_{i_0,k}^{(1)}\right)-\phi_{i_0}^{(2)}\left(x_{i_0,k}^{(2)}\right)\|\\&\le \sqrt{d}L_{3,i_0}\|\Delta x_{i_0,k}\|\\&= \sqrt{d}\delta L_{3,i_0}\end{aligned}\tag{32}$$

where the inequality $\|x\|_1\le(d)^{1/2}\|x\|$ for $x\in\mathbb{R}^d$ is used in the first inequality.

Similarly, according to (19) and Assumption 2, one has

$$\begin{aligned}&\|\Delta\xi_{i_0,k}\|_1 = \|\Delta y_{i_0,k}\|_1\\&\le \sqrt{d}\Big\|\left(\nabla_2 f_{i_0}^{(1)}\left(x_{i_0,k}^{(1)},\sigma_{i_0,k}^{(1)}\right)-\nabla_2 f_{i_0}^{(2)}\left(x_{i_0,k}^{(2)},\sigma_{i_0,k}^{(2)}\right)\right)\\&\quad-\left(\nabla_2 f_{i_0}^{(1)}\left(x_{i_0,k-1}^{(1)},\sigma_{i_0,k-1}^{(1)}\right)-\nabla_2 f_{i_0}^{(2)}\left(x_{i_0,k-1}^{(2)},\sigma_{i_0,k-1}^{(2)}\right)\right)\\&\quad-\left(\xi_{i_0,k-1}^{(1)}-\xi_{i_0,k-1}^{(2)}\right)\Big\|\\&= \sqrt{d}\|\nabla_2 f_{i_0}^{(2)}\left(x_{i_0,k}^{(1)},\sigma_{i_0,k}^{(1)}\right)-\nabla_2 f_{i_0}^{(2)}\left(x_{i_0,k}^{(2)},\sigma_{i_0,k}^{(2)}\right)\|\\&\le \sqrt{d}L_{2,i_0}\left(\|\Delta x_{i_0,k}\|+\|\Delta\sigma_{i_0,k}\|\right)\\&\le \sqrt{d}\delta L_{2,i_0}\left(1+L_{3,i_0}\right).\end{aligned}\tag{33}$$

From (30), and substituting (32) and (33) into (31) yields

$$\begin{aligned}&\frac{P(\{\boldsymbol{\eta},\boldsymbol{\xi}\}\in\Omega|\mathcal{M}_{\mathcal{F}^{(1)}}(\boldsymbol{\eta},\boldsymbol{\xi})\in\mathcal{H})}{P(\{\boldsymbol{\eta},\boldsymbol{\xi}\}\in\Omega|\mathcal{M}_{\mathcal{F}^{(2)}}(\boldsymbol{\eta},\boldsymbol{\xi})\in\mathcal{H})}\\&\le\exp\left(K\sqrt{d}\delta\frac{L_{3,i_0}+L_{2,i_0}(1+\sqrt{d}L_{3,i_0})}{\theta_{i_0}}\right)\\&\le e^{\epsilon_{i_0}}\end{aligned}\tag{34}$$

which establishes ϵ_{i_0}-DP of agent i_0. Since i_0 is arbitrary, the result holds for all agents $i\in\mathcal{V}$, thereby completing the proof. ■

Remark 4: Although the aggregative term introduces coupling among agents, the sensitivity induced by a single-agent data change remains localized and is masked prior to information exchange. Due to the postprocessing property of DP, the consensus-based mixing does not increase the effective sensitivity of shared messages. Hence, the proposed noise calibration remains valid in the distributed aggregative setting.

From Theorem 1, it can be observed that the upper bound in (21) is a function of α, β, $\bar{\theta}$, and other parameters. Notably, the bound decreases as $\bar{\theta}$ decays. When the system parameters are fixed, the tradeoff between privacy level $\min_i\epsilon_i$ and optimization accuracy ϱ is characterized by the following asymptotic relation:

$$\varrho\sim O\left(1/\min_i\epsilon_i\right)\tag{35}$$

for small ϵ_i. This implies that as $\epsilon_i\to 0$, i.e., in the case of complete privacy for each agent, the optimization accuracy becomes arbitrarily low.

VI. Numerical Simulations

In this section, we validate the effectiveness of our proposed Algorithm 1 through numerical simulations.

Case I (The Optimal Placement Problem): We consider a representative optimal facility location problem in $\mathbb{R}^2$. Specifically, there are five fixed entities ($M = 5$) at predetermined locations $r_1 = (3,5)$, $r_2 = (6,9)$, $r_3 = (9,8)$, $r_4 = (6,2)$, and $r_5 = (9,2)$, and five free entities ($N = 5$) whose positions need to be determined. Each free entity can only privately access a subset of the fixed entities. The objective is to determine the optimal positions for the free entities to minimize the total distance between each free entity and its corresponding fixed entity, as well as the distance from each free entity to the weighted centroid of all free entities. For example, the free entities may represent transfer stations where goods are first delivered to these stations, and subsequently distributed from the stations to the fixed locations.

For each free entity $i\in\{1,2,\ldots,5\}$, the local objective function is defined as

$$f_i\left(x_i,\sigma\left(\boldsymbol{x}\right)\right)=\bar{r}_i\|x_i-r_i\|^2+\|x_i-\sigma\left(\boldsymbol{x}\right)\|^2\tag{36}$$

where $\bar{r}_i = i+1$ is a weighting parameter, and $\sigma(\boldsymbol{x}) = (1/5)\sum_{i=1}^{5}x_i$ is the average position of all free entities. $x_i\in\mathbb{R}^2$ is the position of free entity i. For simulation, we set $\alpha = 0.02$, $\beta = 0.3$, and define ϕ_i as the identity mapping with respect to i. To quantify the privacy guarantees, the benchmark constants are specified as $\delta = 1.0$ and a finite horizon of $K = 100$ iterations. Under this setting, a baseline noise scale $\theta_i = 0.01$ is adopted for the convergence analysis.

First, based on the objective formulation in (36), we apply the proposed Algorithm 1 to solve the DAO defined in (1).

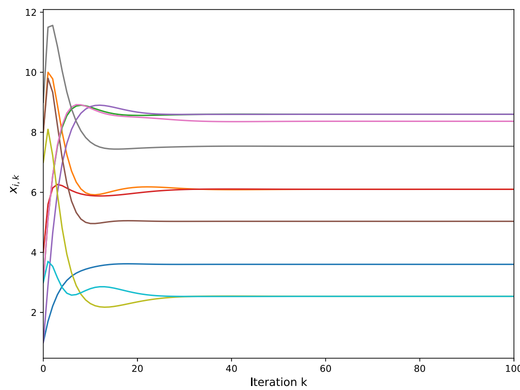


Fig. 1. Evolutions of $x_{i,k}$. Approximate x_i by averaging over more than 50 simulation results.

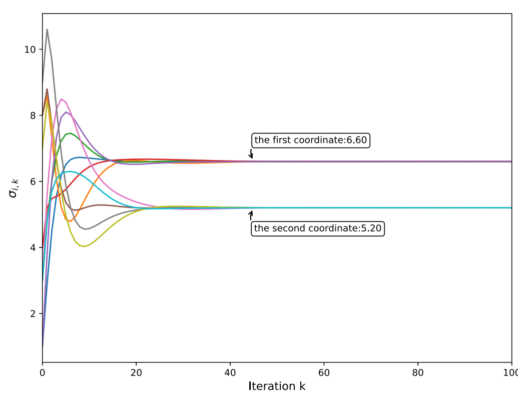


Fig. 2. Evolutions of $\sigma_{i,k}$. Approximate σ by averaging over more than 50 simulation results.

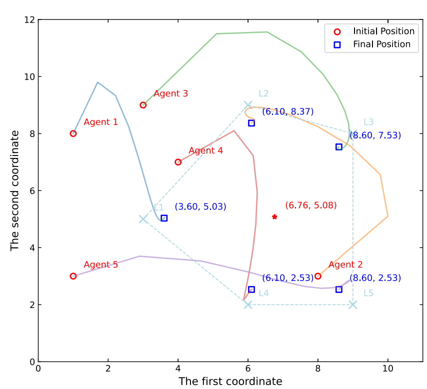


Fig. 3. Evolutions of $x_{i,k}$. Approximate x_i by averaging over more than 50 simulation results.

The evolution of the free entities' decision variables x_i and the aggregated variable $\sigma(\boldsymbol{x})$ is depicted in Figs. 1 and 2, respectively. As shown in Fig. 3, each free entity converges to its corresponding suboptimal position, while the local estimates of the aggregated variable σ_i achieve consensus at the unified point. These results confirm the theoretical guarantees and validate the effectiveness of the proposed algorithm design.

Second, Fig. 4 evaluates the convergence performance of four algorithms using the objective error gap $|F(\boldsymbol{x}_k) - F^*|$. Under identical noise conditions, the proposed PP-HB-DAGT converges to a neighborhood of the optimal solution, whereas the traditional DAGT exhibits unstable behavior and

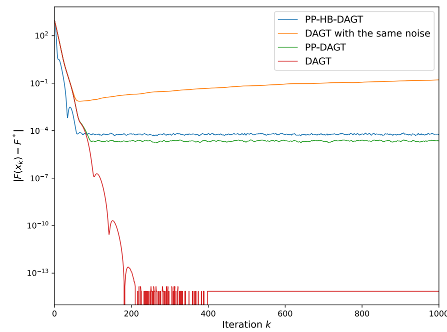
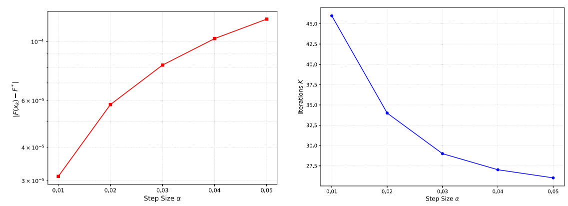
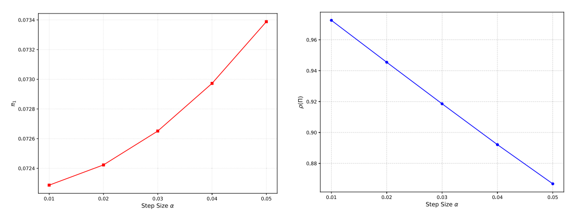


Fig. 4. Evolutions of the objective error $|F(x_k)-F^*|$ under different algorithms. The expected residual is approximated by averaging over 50 simulation results.



(a) (b)

Fig. 5. Comparison of accuracy and convergence speed across different step sizes, averaged over 50 independent simulations. (a) Accuracy versus step size. (b) Convergence speed versus step size.



(a) (b)

Fig. 6. Theoretical performance bound curves of π_1 and the spectral radius $\rho(\Pi)$ versus parameter α. (a) Optimality gap versus step size. (b) Spectral radius $\rho(\Pi)$ versus step size.

fails to attain convergence. As a baseline, the noise-free DAGT achieves global optimality. Furthermore, a comparison between PP-HB-DAGT and its nonmomentum counterpart (PP-DAGT) reveals an inherent tradeoff: while the momentum term accelerates the transient convergence, it slightly increases the steady-state error, leading to a minor degradation in final accuracy.

Third, to further investigate the impact of the step size α on algorithm performance and to validate the theoretical findings in Theorem 1, we examine the convergence accuracy and convergence speed under varying values of α. The results are presented in Fig. 5, while the corresponding theoretical metrics are plotted in Fig. 6. Fig. 5(a) demonstrates that the steady-state objective error increases with α, which is consistent with the theoretical error bound coefficient π_1 shown in Fig. 6(a). In contrast, Fig. 5(b) shows that a larger α reduces the average number of iterations K required to reach the convergence threshold, indicating a faster convergence rate. This trend is theoretically supported by Fig. 6(b), where the spectral radius $\rho(\Pi)$ decreases as α increases. Together, these results confirm the theoretical tradeoff between convergence speed and steady-state accuracy and motivate the choice of α in the simulations.

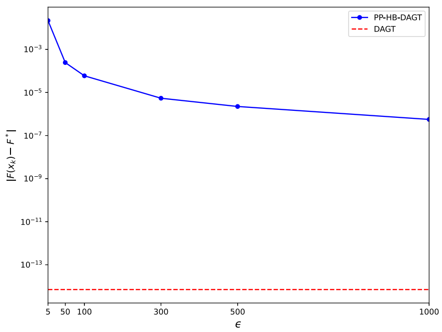


Fig. 7. Impact of the privacy budget ϵ on the steady-state objective error $|F(x_k) - F^*|$, averaged over 50 independent simulations.

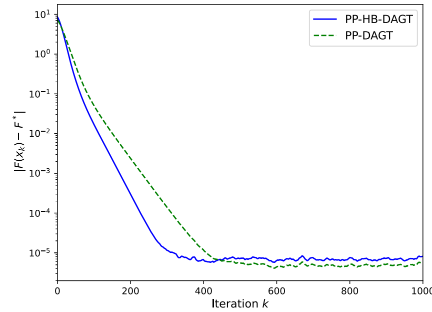


Fig. 8. Evolutions of the objective error $|F(x_k) - F^*|$ under different algorithms. The expected residual is approximated by averaging over 50 simulation results.

Fourth, to empirically validate the asymptotic tradeoff established in (35) and to explicitly quantify the impact of the privacy budget on algorithm performance, Fig. 7 illustrates the steady-state objective error $|F(x_k) - F^*|$ against varying privacy budgets ϵ. The performance of the nonprivate DAGT algorithm is plotted as a horizontal dashed line to serve as a benchmark. As observed in Fig. 7, the steady-state objective error decreases monotonically as ϵ increases, which is consistent with the orderwise dependence on ϵ derived in (36). Specifically, in the high-privacy regime (small ϵ), the algorithm exhibits a larger steady-state error due to the increased noise scale required to satisfy stricter DP constraints. Conversely, in the high-budget regime (large ϵ), the injected noise becomes negligible and the performance of the proposed PP-HB-DAGT algorithm gradually approaches that of the nonprivate baseline. This visualization confirms that ϵ serves as a critical tuning parameter that governs the tradeoff between privacy protection and optimization accuracy.

Case II (Distributed Logistic Regression With Regularization): To evaluate the performance of the proposed algorithm on nonquadratic DAO problems, we consider a distributed logistic regression task with consensus regularization. The local objective function for agent i is defined as $f_i(x_i, \sigma(\boldsymbol{x})) = (1/m_i)\sum_{j=1}^{m_i}(\ln(1 + e^{\mathbf{c}_j^T x_i}) - y_j(\mathbf{c}_j^T x_i)) + (\lambda/2)\|x_i - \sigma(\boldsymbol{x})\|^2$, where $\mathbf{c}_j \in \mathbb{R}^d$ contains $d = 2$ features of the training data and $y_j \in \{0, 1\}$ is the corresponding observed label. The second term serves as a consensus regularization with $\lambda = 0.5$. Each agent i has $m_i = 50$ training data. The feature vector $\mathbf{c}_j$ is generated independently over i, and each entry is drawn from an i.i.d Gaussian distribution with zero mean and variance 10. The true parameter vector $x \in \mathbb{R}^d$ is drawn from a standard normal distribution. The label y_j is generated from a Bernoulli distribution with a probability of $y_j = 1$ being $1/[1 + e^{-\mathbf{c}_j^T x}]$.

To explicitly disentangle the impact of the heavy-ball momentum term on the convergence rate versus the steady-state error, Fig. 8 presents a comparative analysis between the proposed PP-HB-DAGT and the nonmomentum baseline, PP-DAGT. The results reveal that PP-HB-DAGT exhibits a steeper descent trajectory compared with PP-DAGT. This confirms that the momentum term effectively accelerates the convergence rate, enabling the algorithm to reach the optimal neighborhood more rapidly. Furthermore, in terms of steady-state behavior, PP-HB-DAGT converges to a slightly larger error neighborhood than PP-DAGT. This phenomenon is consistent with the theoretical bound in Theorem 1, where the steady-state error coefficient π_1 depends explicitly on the momentum parameter β.

VII. Conclusion

This article considers the problem of privacy-preserving accelerated DAO. To address this problem efficiently, we propose a novel algorithm, termed PP-HB-DAGT, which integrates DAGT with heavy-ball momentum, DP, and a fixed step size. We rigorously establish that the proposed algorithm achieves a linear convergence rate under the assumption that the global objective function is strongly convex and has Lipschitz-continuous gradient. Furthermore, we reveal that the privacy parameter ϵ governs a fundamental tradeoff between privacy levels and optimization accuracy. Numerical experiments validate the effectiveness of the findings. Future work will focus on enhancing optimization accuracy and addressing constrained distributed aggregated optimization problems.

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